

Amlodipine Besylate API - Commercial Plant Handover

Document Type: Quality Assurance (QA) Batch Disposition & Plant Compliance Dossier
Plant Location: Site 4, Unit-B (Commercial Manufacturing Facility)

Target Drug API:	Amlodipine Besylate	QA Protocol ID:	QA-AML-2026-COMP-01
Manufacturing Site:	Unit-B, Block 3 (Rx-04)	Target Markets:	US FDA (PAS) / EU (Type II)
QA Review Lead:	H. Vance, Senior QA Director	Review Period:	Q1 2026 Batch Releases

1. Quality Assurance Executive Summary & Audit Scope

This dossier represents the formal Quality Assurance execution package for the commercial-scale validation batches of Amlodipine Besylate API synthesized via the amended synthetic pathway (KSM-3 introduction). The QA department has evaluated executed Batch Manufacturing Records (eBMRs), Quality Control Certificate of Analysis (CoA) records, plant deviation logs, and equipment qualification metrics to determine final site disposition.

QA Review Area	Batches Assessed	QA Compliance Status	Action Required
eBMR Review	AML-2026-01, 02, 03	COMPLIANT	Sign-off for Release
OOS / OOT Investigations	AML-2026-02	OPEN INVESTIGATION	Hold Batch Release
Nitrosamine Testing	AML-2026-03	FLAGGED	Follow-up CAPA
Process Validation	AML-2026-01 to 03	UNDER REVIEW	Re-evaluation Required

2. Executed Batch Manufacturing Record (eBMR) Audit

QA conducted a page-by-page audit of executed BMRs for 3 commercial validation batches (500 kg scale). Parameters were verified against Master Batch Records (MBR-AML-REV04).

Step / Unit Op	Master Limit	Batch 01 (eBMR)	Batch 02 (eBMR)	Batch 03 (eBMR)
Step 1 Reflux Temp	70°C - 78°C	75.2°C	76.0°C	74.8°C
Step 1 Agitation	120 ± 10 RPM	122 RPM	118 RPM	120 RPM
Step 2 Amine Addition	35°C - 45°C	40.1°C	39.8°C	41.2°C
Step 2 Deprotection Time	6.0 Hours Max	6.0 Hours	8.5 Hours	5.8 Hours
Step 3 Acid Addition	90 - 150 Min	120 Min	110 Min	125 Min

Observation notes recorded during eBMR review indicate extended dwell time during Step 2 deprotection for Batch AML-2026-02 attributable to operational handover during shift rotation.

3. Finished Product Certificate of Analysis (CoA) & Analytical Records

Analytical release testing results from QC LIMS were audited against finished product release specification STP-QC-AML-01.

Analytical Test	Specification Limit	Batch 01 Result	Batch 02 Result	Batch 03 Result
Assay (HPLC)	98.0% - 102.0%	99.6%	97.1%	99.5%
Impurity A	NMT 0.15%	0.04%	0.28%	0.05%
Impurity F	NMT 0.15%	0.05%	0.06%	0.04%
Residual IPA	NMT 5000 ppm	380 ppm	410 ppm	420 ppm
Water Content (KF)	NMT 0.5%	0.18%	0.22%	0.19%

Analytical analytical values for Batch AML-2026-02 reflect secondary degradation product elevation. Phase II analytical re-testing protocol was initiated by QC.

4. ICH M7 Nitrosamine Testing & Analytical Trending

Confirmatory testing via LC-MS/MS was performed for trace N-Nitrosopiperidine (NPIP) and N-Nitroso-amlodipine (NDSRI).

Impurity Target	AI Limit (PPM)	Batch 01 Level	Batch 02 Level	Batch 03 Level
N-Nitrosopiperidine	130 ppm	< 0.02 ppm	< 0.02 ppm	0.01 ppm
N-Nitroso-amlodipine	150 ppm	< 0.02 ppm	< 0.02 ppm	1.25 ppm
Benzenesulfonate Esters	10 ppm	< 0.10 ppm	< 0.10 ppm	< 0.10 ppm

Historical baseline profile comparison highlights variation in NDSRI response for Batch AML-2026-03. Material submitted for exploratory structural confirmation.

5. Commercial Process Validation & Yield Recovery Analysis

Comparative evaluation of 3 consecutive commercial batches against validation protocol target yields (PVP-AML-2026).

Validation Stage	Target Yield Scope	Batch 01	Batch 02	Batch 03
Step 1 Cyclization	87.0% - 90.0%	88.5%	88.2%	88.8%
Step 2 Deprotection	90.0% - 93.0%	91.8%	74.1%	91.5%
Step 3 Salt Formation	92.0% - 95.0%	93.8%	90.2%	94.0%
Overall Process Yield	75.0% - 78.0%	76.2%	58.9%	76.4%

Batch AML-2026-02 yield profile evaluated under validation campaign metrics. Step 2 efficiency loss contributed to lower material recovery.

6. Plant Deviation, CAPA & Quality Record Summary

Consolidated tracking table of quality events logged during the validation batch execution run.

Record ID	Affected Batch	Event Description	Severity	Status
DEV-2026-011	AML-2026-02	Step 2 Extended Deprotection Time	MAJOR	OPEN
OOS-2026-004	AML-2026-02	Assay & Impurity A Deviation	CRITICAL	UNDER INVESTIGATION
OOT-2026-002	AML-2026-03	NDSRI Elevation Trend	MAJOR	PENDING CAPA
VAL-2026-001	AML-2026-02	Stage Yield Variance	CRITICAL	OPEN

7. QA Disposition Sign-off Block

Final disposition record for commercial scale validation batches.

QA Function	Reviewer Name	Disposition Decision	Date / Signature
QA Compliance Lead	S. Jenkins	REJECT Batch AML-2026-02	28-Jan-2026 [SIGNED]
VP Quality Operations	Dr. M. Vance	HOLD Regulatory Filing	28-Jan-2026 [SIGNED]